

RAKSHITH MATHAD

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EDUCATION

University of Southern California, Los Angeles, California - Master of Science in Applied Data Science

AUG 2022-MAY 2024

Coursework Highlights: Machine Learning for Data Science and AI, Data Mining, Predictive Analytics, Fairness, Security and Privacy in AI

KLE Technological University, India - Bachelor of Engineering in Computer Science

AUG 2018-JUN 2022

Coursework Highlights: Data Structures and Algorithms, Data Mining, Computer Architecture, Operating Systems, Distributed & HPC, Cloud Computing

SKILLS

Languages Python, SQL, C, C++, CUDA(Intermediate)

Tools & Technologies HW/SW co-design , GCP Vertex AI, AWS, Docker, Distributed Systems, Git, Informatica ETL, Parallel Programming, GCP ADK, VectorDBs, LLM Training and Inference Optimization, LoRA SFT, HuggingFace, FastAPI, REST APIs, Linux, Pandas, Numpy, Scikit-learn, MongoDB, Hadoop HDFS, PySpark, PowerBI/Tableau, Redis, TensorRT, ONNX, LLMs, Generative AI, Azure, Google Cloud Platform, BigQuery, Kubernetes, Horovod, Jenkins CI/CD, NLP, Deep Learning, Machine Learning, A/B Testing, Grafana

Certifications NVIDIA CUDA Computing ,Juniper Networks Certified Associate, TensorFlow/PyTorch for Artificial Intelligence, Agile, TCP/IP

EXPERIENCE

CVS Health (Aetna Insurance) - Software Engineer – AI/ML, New York City, NY

JUN 2024- PRESENT

- As a generalist **Solutions Engineer**, I work in the **AI call center customer facing** team to build a large-scale high volume low latency RAG and Rule based **FastAPI** chat application using **GCP Google AI Platform**. My contributions to the project has an **impact of ~ \$12 MM/yr and I received a company-wide award for high and vibrancy impact**. Worked with **OpenAI Throughput Units and Gemini LLMs**, employed **Feature Store** semantic search engine to build async semantic search based information retrieval and answer generation modules, resulting in improved latency and caching strategy. Helped in profiling and **scaling this production AI system to ~20,000 req/hr impacting around 200k customers everyday** leveraging **Kubernetes GKE, Vertex AI** and maintaining the **Kubernetes infrastructure including GPUs**.
- Performed complex data preprocessing on GCP BigQuery and BigTable, including chunking and embedding strategies. Took part in design reviews, **building MCP agents**, improved latency by **1.5x** using **TensorRT/ONNX model optimizations** for the embedding service.
- Also worked with Airflow, Salesforce CRM, **K8s**, Jenkins CI/CD, LLM Evaluation and regression framework, scraping/automation, Redis, feature flags, On call rotations with prod deployments, advanced prompt engineering, load testing, AI guardrails etc. I constantly produced multiple impactful PRs every week.

CVS Health - Analytics Engineering Intern, New York City, NY

MAY 2023- AUG 2023

- Optimized Hadoop HDFS big data pipelines, processing 100M+ rows with HiveQL, cutting latency by 40%. Designed scalable ETL workflows and **boosted customer campaign growth** strategy leveraging **Tableau**, advanced OLAP, data blending, and predictive modeling.
- Built complex **Azure** Data Factory pipelines for large data processing, used **PySpark** in **Databricks** for batch API processing and Dynamics CRM data migration via Python OData REST APIs. Handled Parquet/Avro for streaming and financial data reporting.

AlphaICs Corporation - AI Software Engineer Intern, India

JAN 2022-JUL 2022

- Developed deep learning applications for Object Detection, Recurrent Models for Visual Attention, Recommendation Systems (Matrix Factorization, Collaborative Filtering) etc for **faster AI Inference on a custom AI processor**.

Samsung R&D Institute - AI Research Intern, India

NOV 2020-JUN 2021

- Led a team of 3, and worked on the SOTA **Generative AI PyTorch** research Implementations for the task of **Semantic Image Editing using Conditional GAN** with spatially adaptive normalization, for Image manipulation with controlled generative-fill. Performed Semantic Segmentation with DeepLab-V2 to curate a 20k-image dataset and trained generative models on an **Nvidia DGX** cluster.

PROJECTS

- Selective EAGLE3 scheduler for vLLM** — routes requests to baseline or **speculative decoding** by prompt length, temperature, and rolling draft acceptance. Validated on Llama-3.1-8B on **H200 GPU**, achieved **+4.2% throughput** vs baseline, skipping spec decode on low-fit traffic.
- Free claude code** - A local proxy server that lets you use Claude Code CLI for free by intercepting its Anthropic API calls and translating them to NVIDIA **NIM's** OpenAI-compatible format, converting requests, responses, and **SSE** streams bidirectionally in real time.
- Voice agent** (VAD → Whisper STT → OpenAI tool-calling LLM → guardrails → TTS) with multi-turn agent harness, SQLite tools, and barge-in.
- Built a **vLLM weight-snapshot** prototype that restores saved weights into warm slots — distilgpt2 activation dropped from **13.5s → 0.74s**; model swaps in **~0.8–1.2s** on GPU. Exploring CRIU + CUDA checkpointing for **serverless cold starts**.
- Built a mini LLM Gateway (**Qwen 3B** served via **SGLang** with AWQ quantization), an Agent Gateway with tool-calling leveraging **MCP**.
- Blogs about **NVIDIA Dynamo serving, etc.** Deployed using BERT Transformer on an RTX GPU in **PyTorch, ONNX, NVIDIA TensorRT, Ray Serve/vLLM** runtimes and profiled it to study performance bottlenecks using **NVIDIA Nsight**. Building small CUDA Kernels(kernel fusion) to understand infusion into vllm/sglang for potential optimized inference.

RESEARCH EXPERIENCE

- Published** a paper "[Performance Analysis of Distributed Deep Learning using Horovod on Image Classification](#)", on **Horovod (used in NCCL) ring all reduce** for multi GPU distributed training on Nvidia DGX cluster at IEEE ICICCS ([Google Scholar](#))
- Machine Learning Engineer (Information Sciences Institue (ISI) – Los Angeles, CA)** – Developed algorithms to leverage ML privacy-preserving techniques to detect anomalies infused by adversary client nodes in a **Federated Learning** system for **Homomorphic Encryption**.